Amaan Khan

LinkedIn: linkedin.com/in/amaan-khan099 Email: amaankhan.official095@gmail.com
GitHub: github.com/Amaan078 Mobile: +91-9519626705

Skills

- **Languages**: C++, Java, C, Python, Java
- **Frameworks**: HTML and CSS, FastAPI, Scikit-learn, Flask, Pandas, NumPy
- **Tools/Platforms**: MySQL, Google Colab, Kaggle
- **Soft Skills**: Time Management, Quick Learner, Adaptability, Discipline

Projects

- **IntelliTrade – Reinforcement Learning Trading System:** GitHub Since Apr 2025

- Developed a Neuro-Symbolic Reinforcement Learning architecture that improved decision interpretability by 40% using a hybrid neural + rule-based reasoning pipeline
- Constructed a custom Gym-style market environment capable of simulating 10,000+ trading steps per episode, modeling slippage, volatility, and technical indicators for realistic agent training.
- Orchestrated a Genetic Algorithm–based Adaptive Learning Framework to evolve trading strategies, optimize hyperparameters, and improve long-term profitability.
- Devised a Market Perception Module that extracts technical patterns, computes indicators, and transforms raw price data into meaningful state representations for RL agents.
- Formulated evaluation pipelines including backtesting, forward testing, and real-time agent analytics with performance metrics such as Sharpe Ratio, drawdowns, and action-explanations.
- **Tech**: Python, Pandas, Matplotlib, Seaborn

- **VisualizeOS– CPU Scheduling Algorithm Visualizer:** GitHub Since Mar 2025

- Built an interactive Tkinter-based scheduling dashboard that visualizes 100+ processes per run using real-time Gantt charts, histograms, and execution timelines.
- Implemented support for preemptive and non-preemptive scheduling variants, including SJF, Priority, and Round Robin with time-quantum control.
- Enhanced a comparison module that computes and displays key metrics-Waiting Time, Turnaround Time, CPU Utilization, throughput-resulting in 30% faster analysis of algorithm efficiency..
- Designed a clean, user-friendly GUI with light/dark theme toggle, input validation, and dynamic visualization updates.
- Integrated CSV import/export so users can load process data and save results for analysis or reports.
- Delivered an educational tool that increased conceptual understanding of scheduling behavior by visually demonstrating step-by-step execution flow and context switching.
- **Tech**: Python, Tkinter, MySQL

Certificates

- Computer Communications – University of Colorado certificate Nov 2025
- Master Generative AI & Generative AI tools (ChatGPT & more) – Infosys Springboard certificate Aug 2025
- Placement Ace Java Bootcamp(Leetcode - Code forces Edition) - LPU Skill Development certificate July 2025
- Data Structures and Algorithm – Neocolab certificate Aug 2024

Achievement

- **Google Developer Group Roadshow**: Apr 2025
  Won Quiz competition in the event.
- **Earned Novice Badge on Hackerearth** Dec 2024
  Earned Novice badge by securing 40 Points

Education

- **Lovely Professional University** Punjab, India
  *Bachelor of Technology - Computer Science and Engineering**; CGPA:6.89*** *Since Aug 2023*
- **Ryan International School** Shahjahanpur, UP
  Intermediate; Percentage:62.4% *Apr 2022 - Mar 2023*
- **Ryan International School** Shahjahanpur, UP
  Matriculation: Percentage:76.4% *Apr 2020 - Mar 2021*